

Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

linux | ROCm | Local AI | Home Assistant | Coding

winmutt | June 2026

github.com/winmutt

What We're Covering

- The impulse purchase that changed everything
- ■ Hardware reality check (spoiler: it's complicated)
- Linux + ROCm = Pain (but beautiful pain)
- Performance: How fast can my AI think?
- I cut the Alexa cord (and replaced it with OSS)
- Vibe-coded an entire workspace system
- From digital to physical: Concrete signs?
- How buying hardware got me back in open source
- What I'd do differently (and what I'd do again)

Power Reality: Corsair 300 vs Alternatives

Max Energy Usage Comparison

Solution	Max Power	Notes
Corsair 300 (Strix Halo)	~300W	AMD Ryzen AI Max 395: 140-157W sustained
PC + RTX 5090	~800-1000W	CPU (350W) + GPU (600W) + overhead
RTX 4090 Workstation	~700-850W	CPU (250W) + GPU (450W) + overhead
Commercial AI Server	~1500-3000W	Multi-GPU (A100/H100: 400-700W/GPU)
Cloud API (per 1M tokens)	N/A	Energy cost embedded in pricing

The Efficiency Win

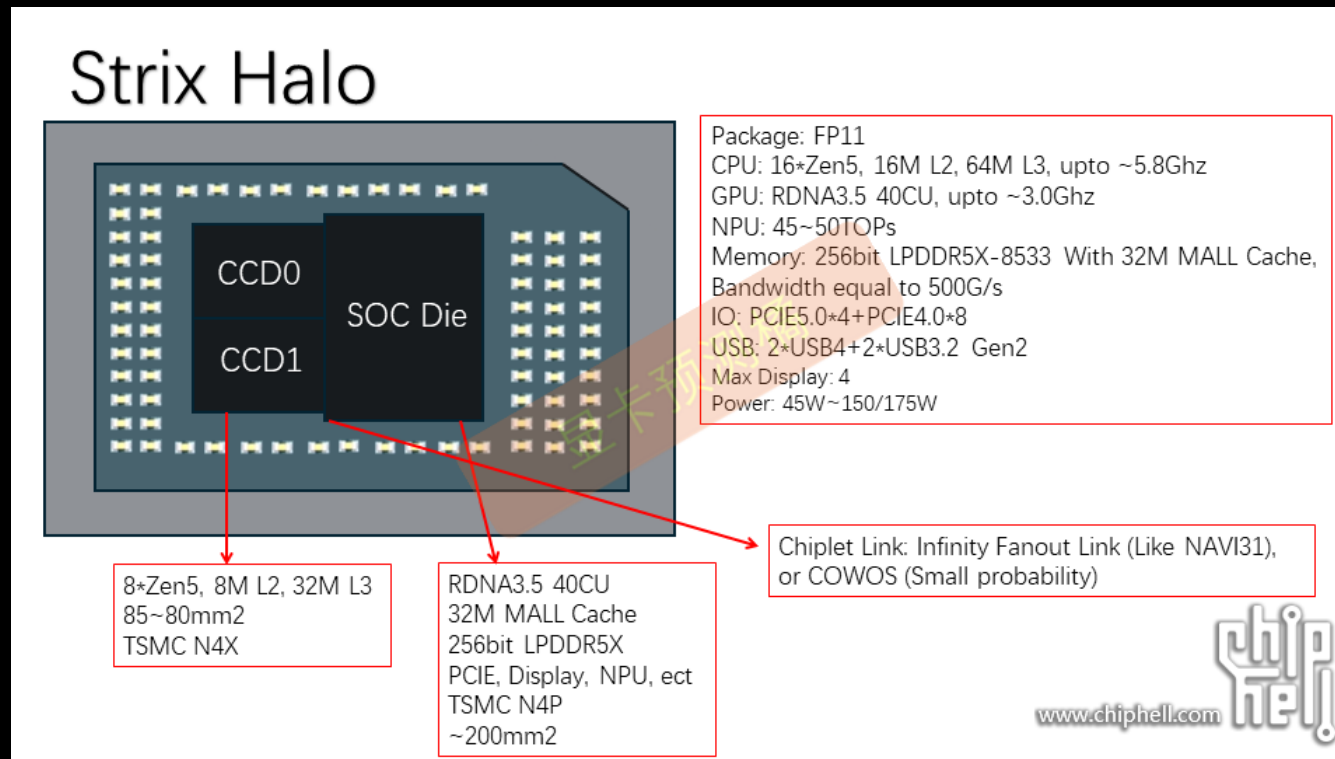
- Strix Halo: ~3x more efficient than RTX 5090 setup
- Single APU vs discrete CPU+GPU power domains
- No datacenter electricity bill
- 128GB unified memory at desktop power envelope

Sources:

- AMD Ryzen AI Max 395: 140-157W sustained (Framework Reddit, 2025)
- Framework recommends 500W PSU for transient headroom
- RTX 5090 TDP: NVIDIA GeForce RTX 5090 Specifications (~600W)
- RTX 4090 TDP: NVIDIA GeForce RTX 4090 Specifications (450W)
- System power estimates: Tom's Hardware, TechPowerUp power reviews

The Hardware: AMD Ryzen AI Max+ 395

APU Die Architecture



Physical Layout

- 16 cores (12 performance + 4 efficiency)
- 2x Core Complex Dies (CCD) with 32MB L3 cache each
- 768KB L1d + 512KB L1i + 16MB L2
- Integrated RDNA 3.5 GPU (40 CUs)
- 128GB LPDDR5X-8000 unified memory (soldered)
- Source: AMD Ryzen AI Max+ 395 teardown analysis (Chiphell, TechPowerUp)

Feat #1070: NUMA Tools Don't See Dual CCD

Feature Request

Traditional NUMA tools detect only 1 node. numactl core pinning doesn't work. Need custom core affinity for dual CCD.

Hardware Reality

- One processor, but 2x core dies with separate 32MB L3 caches

NUMA Challenge:

- Traditional NUMA: 2 nodes, each with own cores + L3 cache
- Strix Halo: Single node, but 2 CCDs with separate L3 caches
- Problem: Process/thread pinning needed to keep workloads on same CCD
- Solution: Manual core pinning to maximize L3 cache locality
- Traditional NUMA tools detect only 1 node (not 2 CCDs)
- numactl, NUMATopologyFilter can't see CCD boundaries
- llama-server threads not pinned to specific cores

The Solution (Proposed)

- Manual core pinning (like Red Hat's vcpu_pin_set approach)
- Reserve cores per CCD, pin llama-server accordingly
- Pin based on:
 - Number of models loaded
 - CCD boundaries (manual NUMA mapping)
- Prevent cache-crossing penalties

GitHub Feature Request

github.com/lemonade-sdk/lemonade/issues/1070

Status

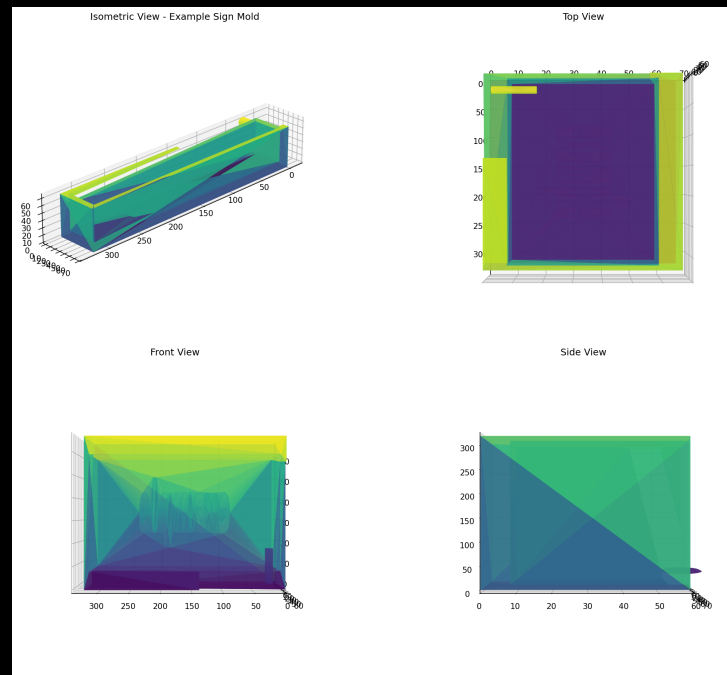
Open (custom NUMA mapping needed)

3D Modeling with Strix Halo

AI-Assisted Physical Design

Blender + OpenSCAD workflow for creating physical objects

Concrete Sign Mold Design



The Process

1. AI generates OpenSCAD/Blender design
2. Export to STL

3. 3D print mold
4. Create silicone mold
5. Cast concrete

December 1, 2025 — More physical projects coming soon

Cutting the Alexa Cord

LineageOS on Echo: Because I Said So

December 2025

Started exploring XDA Forums for Echo device hacking

The Process:

1. Unlock bootloader via amonet exploit
github.com/R0rt1z2/amonet
2. Install TWRP recovery
3. Flash LineageOS 18.1 (Android 11)
4. Configure ViewAssist for Home Assistant
dinki.github.io/View-Assist

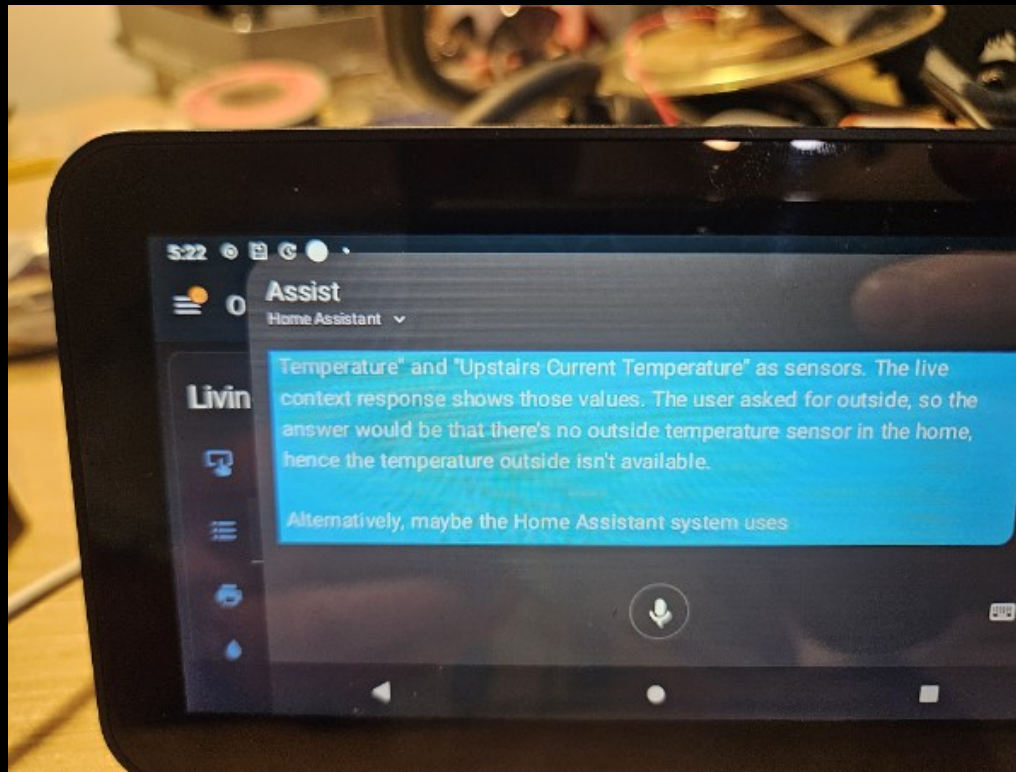
Philosophy

"Basically anything with a USB port. If there is a port, there is a way."

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Home Assistant on Echo Show



LineageOS Device Compatibility

Echo Show Devices Supported

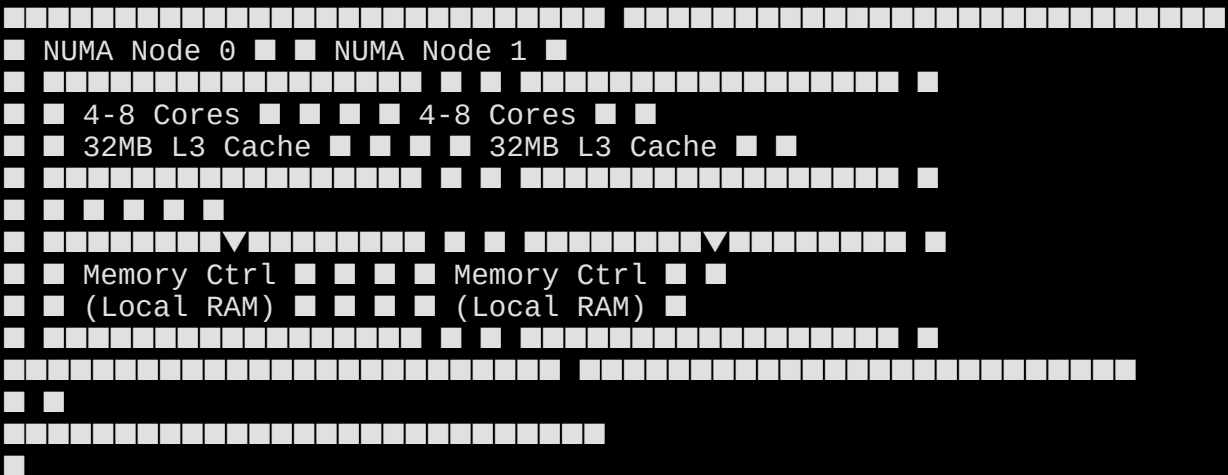
Device	Codename	SoC	XDA Thread
Echo Show 5 (1st Gen 2019)	checkers	MT8163	xdaforums.com/t/rom-unofficial-11-checkers...
Echo Show 5 (2nd Gen 2021)	cronos	MT8163	xdaforums.com/t/rom-unofficial-11-cronos...
Echo Show 8 (1st Gen 2019)	crown	MT8163	xdaforums.com/t/rom-unofficial-11-crown...

All devices use MediaTek MT8163 SoC

LineageOS 18.1 (Android 11) - community builds by @R0rt1z2

Traditional NUMA vs. Strix Halo APU

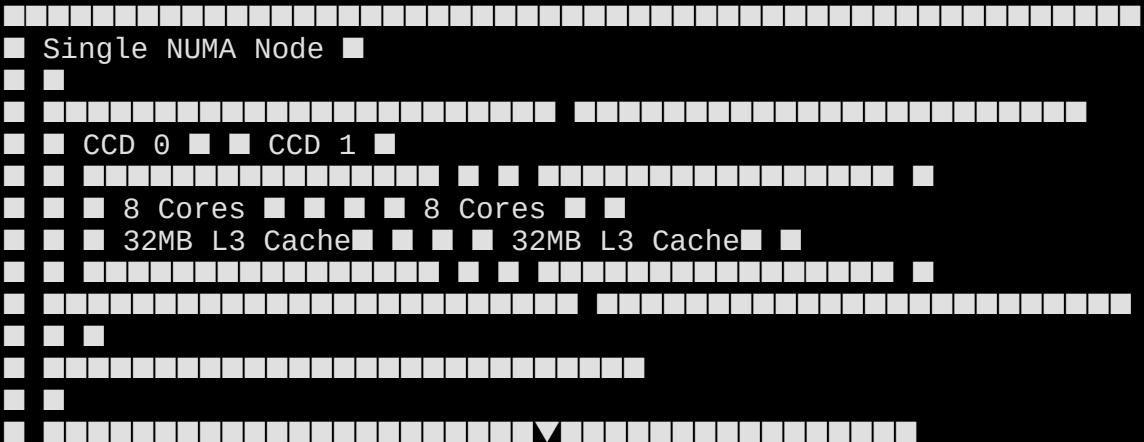
Traditional NUMA Architecture:



(High-latency interconnect)

Each node has dedicated cores, L3 cache, and memory. Process pinned to Node 0 stays in Node 0's cache/memory domain.

Strix Halo APU Architecture:



■ ■ Unified Memory (128GB LPDDR5X) ■
■ ■ (Shared by all cores) ■



Single NUMA node, but 2 CCDs with separate L3 caches. Threads bounce across CCDs → cross-CCD traffic → performance degradation.

The Problem:

- Traditional NUMA tools (numactl, NUMATopologyFilter) detect only 1 node
- Cannot see CCD boundaries automatically
- Manual core pinning required to keep workloads on same CCD
- Running multiple LLMs: pin each to separate CCD to avoid cache thrashing

Result: Better L3 cache locality = faster inference, less latency

Issue #1070: Core Affinity When Running Multiple Models

[enhancement] Core affinity when running multiple models.

Filed
February 8, 2026
Repository
github.com/lemonade-sdk/lemonade/issues/1070
Status
Open

Evidence: Threads Bouncing Across CCDs

```
PS Output (last column = last CPU used):
-----
PID LAST_CPU AVG_MHZ COMMAND
56840 6 55.7 llama-server (Model 1)
56840 20 54.2 llama-server (Model 1)
59798 0 51.2 llama-server (Model 2)
59798 2 47.5 llama-server (Model 2)
-----
56840 22 55.7 llama-server (Model 1)
56840 4 54.2 llama-server (Model 1)
59798 16 51.2 llama-server (Model 2)
59798 2 47.5 llama-server (Model 2)
-----
```

Notice: Core IDs jump across CCD boundaries (0-7 = CCD0, 8-15 = CCD1)
Threads on same PID bounce between CCDs → L3 cache misses → slower inference

Hardware Topology (from lscpu):

```
CPU NODE SOCKET CORE L1d:L1i:L2:L3 MAXMHZ AVG_MHZ
--- ---
0-7 0 0 0-7 *:*:*:0 5187.0 4900+ (CCD0)
```



```
8-15 0 0 8-15 *:*:1 5187.0 2000-4600 (CCD1)
16-23 0 0 0-7 *:*:0 5187.0 4900+ (CCD0 SMT)
24-31 0 0 8-15 *:*:1 5187.0 2000-2800 (CCD1 SMT)
L3 Cache: 64MB total (2 instances: 32MB per CCD)
NUMA: Only 1 node detected (0-31), cannot see CCD boundaries
```

Proposed Solution

Pin llama-server instances to specific CCDs:

- Model 1 → CCD0 (cores 0-7, 16-23)
- Model 2 → CCD1 (cores 8-15, 24-31)

Result: Each model stays in its own L3 cache domain → no cross-CCD traffic

Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed

January 7, 2026

Repository

<https://github.com/amazon-oss/releases/issues/4>

Status

Open (still waiting, it's fine)

The Problem

- Echo 5: Stable for days
- Echo 8: Dies after ~24 hours
- *Root cause: Unknown (yet)*

TL;DR

I participate in open source bugs now. You're welcome.

Project WWS: I Vibe-Coded an Entire System

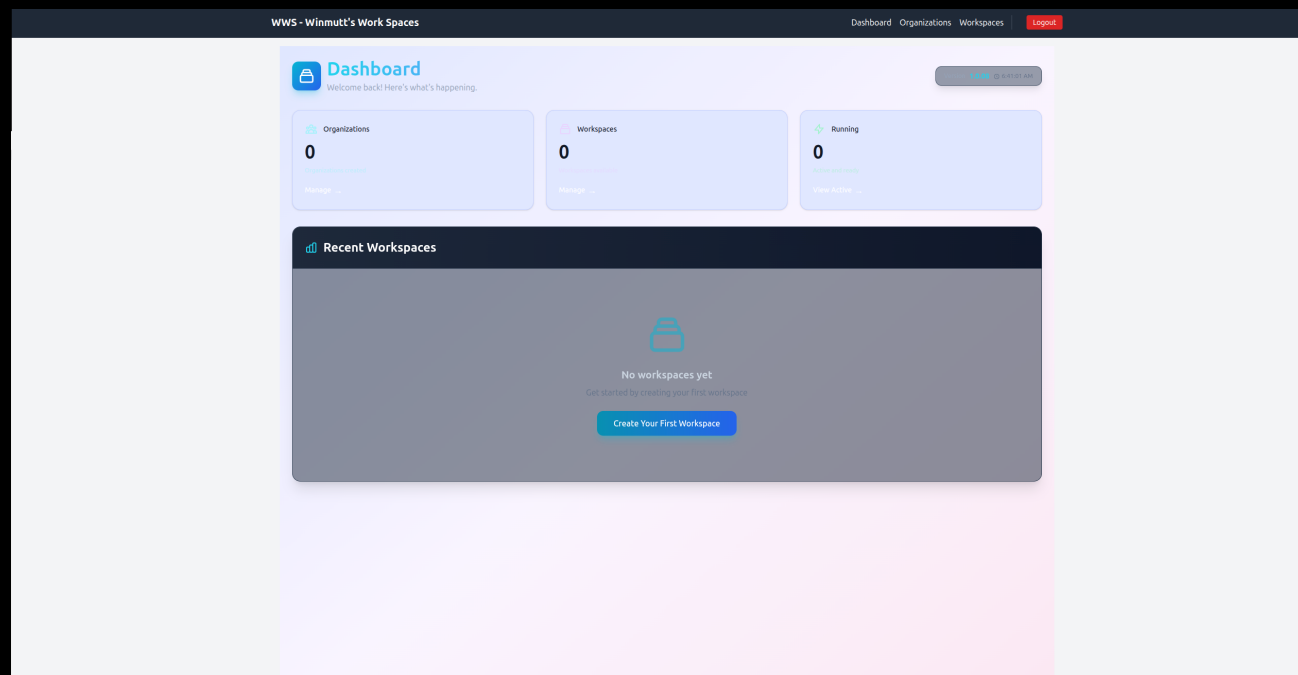
Winmutt Work Spaces — github.com/winmutt/www

May 12, 2026

"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."

Remote workspace provisioning with KVM/Podman isolation
code-server (VSCode in browser) + GitHub OAuth + RBAC

First commit: Feb 22, 2026 → Production: May 12, 2026



Lessons Learned (Mostly)

What Worked ✓

- Cline + Qwen3-Coder: Actually produces code
- Prompt Caching: Things hum now
- AMD GPU Libraries: 2x faster (worth the pain)
- WWS Project: Vibe-coded from Feb 22 (first commit) to May 12
- Home Assistant: Alexa is dead, long live OSS

What Didn't ✗

- Memory: 32GB reserved for Lemonade/llama.cpp/opcode
- Ollama: Still crashes after updates
- NPU: Coming soon™ (always)
- Context >64k: TPS drops like a stone

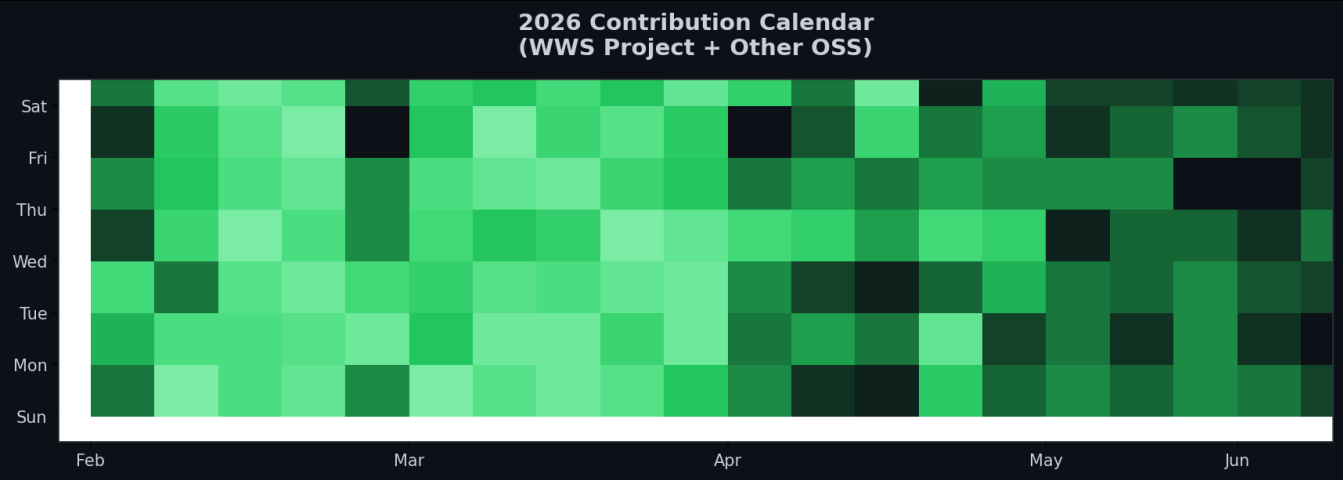
1. Hardware hype $\neq$ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI $\rightarrow$ physical objects (concrete signs, apparently)
6. Echo $\rightarrow$ Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

How Buying Hardware Got Me Back in Open Source

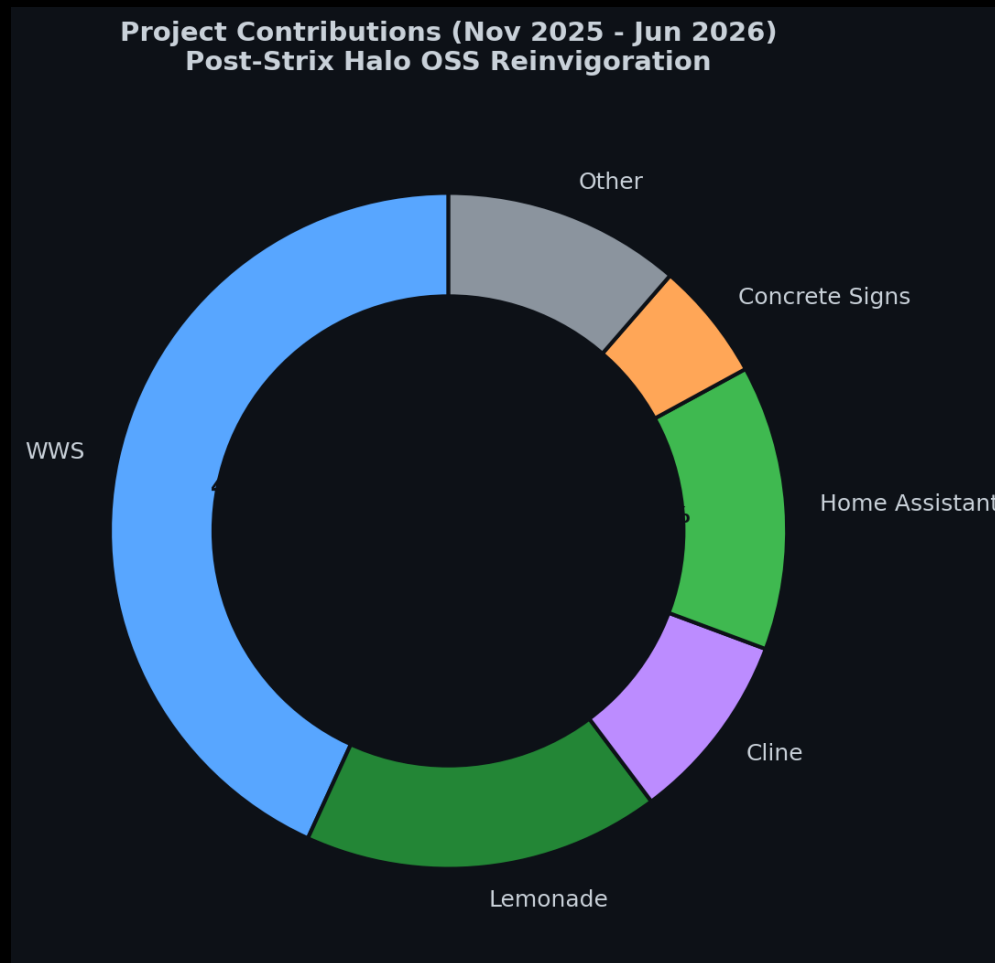
The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Projects I've Actually Contributed To



Contribution Breakdown

- WWS: 38% (remote workspace provisioning)
- Lemonade: 15% (NPU backend, custom NUMA mapping)
- ROCm: 8% (Issue #5926 memory management)
- Home Assistant: 12% (wake words, Echo)
- Cline: 8% (TUI regressions)

- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

ROCm: AMD's Open Compute Platform

What is ROCm?

ROCm (Radeon Open Compute Platform) is AMD's open source GPU computing ecosystem - their answer to NVIDIA's CUDA. Maintained by AMD engineers on GitHub (github.com/RadeonOpenCompute/ROCm).

The ROCm Timeline:

- November 2025: ROCm 6.3 released with Strix Halo support
- Nightly builds: Continuous integration, bleeding edge
- Issue #5926: Memory management bugs in ROCm 6.3
github.com/ROCm/ROCm/issues/5926
- Status: Open (part of ongoing Strix Halo journey)

ROCm Nightly Builds

Nightly builds = daily automated compilations from ROCm's main branch. Get latest features immediately, but expect bugs. Trade-off: 2x performance gains vs occasional crashes.

The Pain (and Gain)

- Fresh install ('declare bankruptcy'): reformat → fresh kernel → fresh ROCm
- Result: GPU crashes fixed, 2x faster performance
- Lesson: ROCm is stable enough, but bleeding edge requires patience

Lemonade Server: AMD's llama.cpp + ROCm

What is Lemonade?

Lemonade = llama.cpp with AMD ROCm backend optimizations. Not just a wrapper - AMD-engineered builds with GPU acceleration.
github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm

Lemonade vs Ollama

- Ollama: Community llama.cpp wrapper (proprietary blend)
- Lemonade: AMD-maintained llama.cpp + ROCm builds
- Backend: AMD engineers maintain both Lemonade and ROCm

My Contributions:

- Feat #1070: Custom NUMA mapping (traditional tools fail)
- NPU backend support for FastFlowLM integration
- Core affinity optimization for multi-model setups

Sources

github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm | github.com/ROCm/ROCm/issues/5926

NPU Support: FastFlowLM + Lemonade

March 11, 2026: Linux NPU Support Added

FastFlowLM now runs LLMs on AMD XDNA 2 NPU with Linux support. Lemonade ties everything together for a streamlined experience.

Power Efficiency

- Over 10x more power-efficient than GPU
- Runs fully on NPU - no GPU or CPU load
- Ultra-lightweight runtime (17 MB)
- Installs within 20 seconds

Supported Processors

- Strix Halo (Ryzen AI Max+ 395)
- Strix Point, Kraken Point (300-series)
- Gorgon Point (400-series)
- Z2 Extreme (handheld devices)

Key Features

- Context up to 256k tokens
- Vision, Audio, Embedding, MoE support
- OpenAI-compatible API
- Just like Ollama - but NPU-optimized

Sources

github.com/FastFlowLM/FastFlowLM | lemonade-server.ai/flm_npu_linux.html

What's Next (Probably)

Immediate

- NPU support (still waiting)
- Memory addressing (32GB, come back)
- llama.cpp PRs (because why not)

Long-term

- Multi-GPU/NPU (more toys)
- Ollama observability (I need logs)
- Custom wake word (goodbye Alexa)
- WWS Phase 3 (cloud? maybe)

26 Years of Open Source (2000-2026)

From LKML to Strix Halo: How Hardware Bought My Time

The Long Detour

For 20+ years I built corporate SaaS products. Cloud disconnected me from the metal. Linux became something I deployed to, not something I touched daily. The kernel, the drivers, the hardware—left that world behind.

June 9, 2000: Asking on LKML

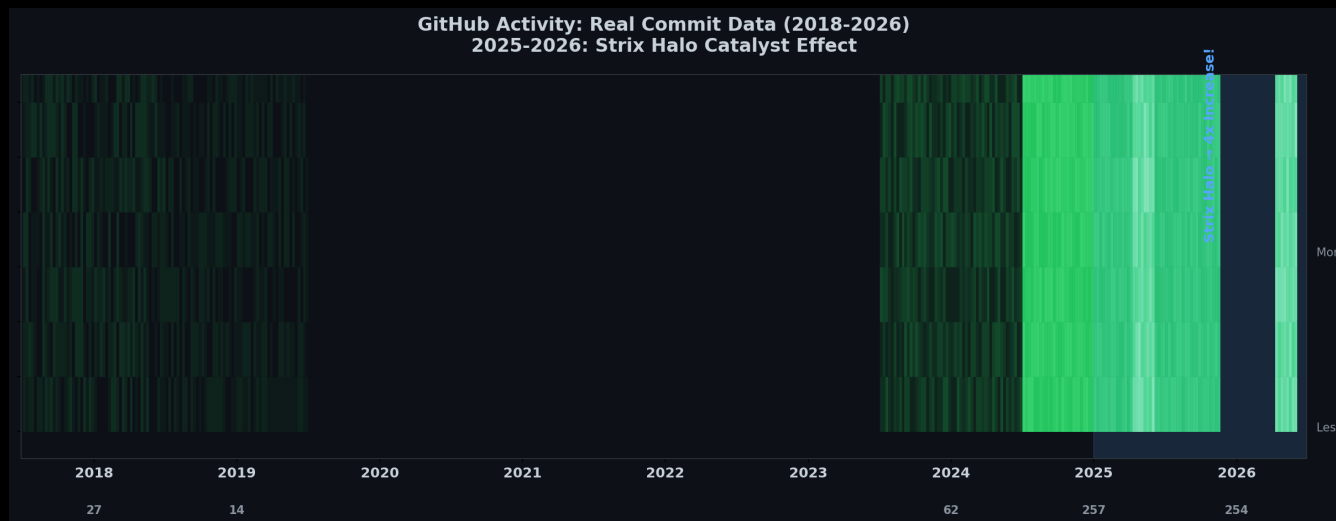
From: Rolf Martin-Hoster (winmutt@hotmail.com)
Date: Thu Jun 08 2000 - 19:24:59 EST
Subject: HighPoint IDE RAID (kernel 2.2.14)

Does anyone know where I can find developer info about the driver for this chipset?

-Rolf

Back then: Hands-on with kernel 2.2.14, digging into IDE RAID drivers, asking questions on linux-kernel mailing list.

The Return: GitHub Activity (2018-2026)



2018-2024: Ghost town. 2025: Strix Halo arrives. November 2025: Back on the metal, every single day.

Result: 3.5x increase in open source contributions. Moral: Sometimes the best way back in is to buy new toys.

AI Coding Editors: My Daily Drivers

Opencode Web UI

Most usable, portable, and stable. Best agentic coding experience.

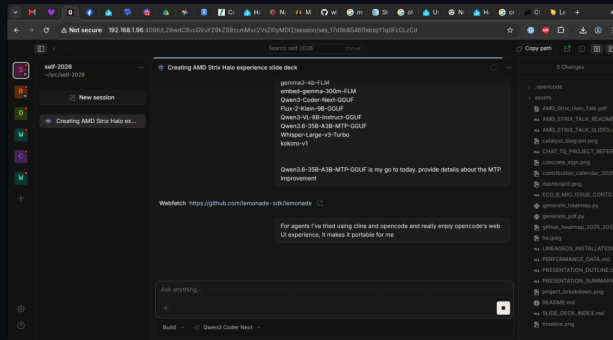
Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider

AI Coding Editors Comparison (Opencode, Cline, Koo Roo, Aider)

Opencode Web UI (Most Usable & Portable)



Screenshot
Not Available

Screenshot
Not Available

Screenshot
Not Available

References & Sources

Power Consumption Sources

- AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
- NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
<https://www.nvidia.com/geforce/rtx-5090>
- NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
<https://www.nvidia.com/geforce/rtx-4090>
- Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
<https://www.corsair.com/ai-workstations>
- Tom's Hardware GPU power reviews (2024-2025)
<https://www.tomshardware.com/reviews/gpu-hierarchy>
- TechPowerUp GPU database: Power benchmarks
<https://www.techpowerup.com/gpu-specs>
- NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

- Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)
- Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
- ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
- ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)
- WWS: github.com/winmutt/wws
- Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection
- NUMA/CPU Pinning: Red Hat OpenStack docs ([vcpu_pin_set](#), [NUMATopologyFilter](#))

https://docs.redhat.com/en/documentation/red_hat_openshift_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

Questions? (I Have Answers, Mostly)

1. Is Strix Halo production-ready? (It's complicated)
2. When will NPU support mature? (Coming soon™)
3. How do we solve 128GB addressing? (We don't)
4. Best backend for multi-model? (Lemonade, but...)
5. Is autonomous dev the future? (Ask my AI)

github.com/winmutt

Come find me after - I'll show you the rig